

Quantification and Depicting Predicates in ASL

Kathryn Davidson and Masashi Tamura

1. Introduction

Human language is often considered to be the paradigmatic case of symbolic abstraction: we can talk about things we've never directly experienced (and perhaps never could), we can associate arbitrary forms to meaning (a rose by any other name would smell just as sweet), and we can combine these symbols into increasingly complex representations which themselves can serve as symbolic input to further computation. Formal syntax and semantics has provided an incredibly productive means by which to model this symbolic system, both its infinite productivity and its constraints.

That said, natural human languages in their communicative context actually involve a spectrum of abstraction. In spoken languages, one productive example of communication with low levels of symbolic abstraction which are still integrated into language are iconic depictive co-speech gestures, which can involve a rather direct mapping between form and meaning. Deictic pointing, onomatopoeia, and other phenomena have long been noted to be closely integrated into human language while still not being fully arbitrary. On the other end of the spectrum, quantifiers like *some*, *none*, *all* involve abstractions over abstractions: they express an abstract symbolic relation between one set (itself an abstraction) and another set (another abstraction). And just as in spoken languages, in sign languages too, we find the entire spectrum of abstraction: we find gesture-like iconic depictions embedded in sign languages via depicting signs (also referred to in sign language linguistics literature as classifier predicates), which involve in part some direct mapping between form and meaning, along with pointing, iconic lexical items, and other non-arbitrary aspects of language. And just as in spoken languages, sign languages also contain highly productive systems of quantifiers, which involve an abstraction over abstractions (a relation between two sets), as illustrated by the negative quantifier *NONE* (Figure 1) which, like its spoken language counterpart, expresses no items of overlap between two sets (here, *P* and *Q*) here from American Sign Language (ASL) (Davidson & Gagne 2022), but well-documented previously for both ASL (Petronio 1995, Abner & Wilbur 2017) and other sign languages such as Catalan sign language (Quer 2012), Russian sign language (Kimmelman 2017), etc.

From the perspective of formal semantic theory for both spoken and signed languages, we might naturally ask if there are any constraints on how the most and least abstract areas of language interact. We know from decades of research on gestures that they can be highly integrated into language production (Kendon 2004, McNeill 2016) but does the meaning contributed by a gesture enter into semantic computations in the same way? Some recent work argues that it does: Tieu et al. (2019) argue that gestures and even pictures are understood by interlocutors to carry the same classes of inferences such as entailments, presuppositions, and implicatures that we observe in language. Others have emphasized instead that iconic gestures seem to contribute meaning on a different dimension of some sorts, and may be more restricted in their compositional potential with symbolic lexical items due to their depictive nature (Kita 1997). One complicating factor involved in studying the question of how iconic content and symbolic content interact in spoken languages is that there is a tendency for iconic depiction to be conveyed using the visual/manual modality (e.g. via gesture) and for abstract symbols to be conveyed using the auditory/oral modality (e.g. via speech), such that there is a trend in which abstraction is confounded with modality difference. This

* Many thanks to our three participants for their time helping us with the project, without which we would not have anything to say here. Sincere thanks also for feedback from the Meaning and Modality Lab, especially our joint collaborator on other projects Nozomi Tomita, as well as collaborators on projects in sign language quantification which form some foundational insights on this question including Deanna Gagne and Annemarie Kocab, and the enthusiastic audience in Seattle at WCCFL in April 2025.

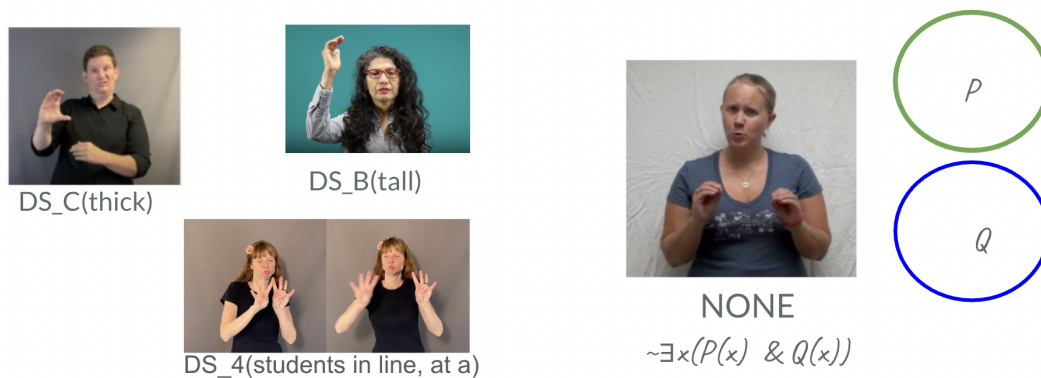


Figure 1: Depicting signs (DS-C, DS-B, and DS-4) involve some of the least abstraction: we understand them through a direct mapping from world to form, while quantification is at the other end of the spectrum, relating two abstractions (i.e. two sets/functions)

has led to the class of ‘gestures’ being in some cases analyzed as having their own semantic type of composition in how they project/can and cannot be targeted by various semantic operators, e.g. “co-supposition” (Schlenker 2018) where the modality difference is taken to be key, but where the iconic/depictive aspects of gesture could instead be playing a critical role.

Toward this end, in this paper we look at two extremes on the spectrum of abstraction in sign languages, where there is an equal variation to spoken languages in terms of kinds of abstraction but where both ends of the spectrum are conveyed in the very same visual/manual modality. In particular, we investigate the interaction of depicting predicates and the way that these interact with quantifiers through three detailed interviews with native signers of American Sign Language targeting productions that involve quantification and depiction. In the following sections we first present some basic linguistic information about depicting signs, and then some basic background on quantifiers. Then, we overview approaches and the predictions they make about the interaction between quantifiers and depicting signs. We then present our data in the form of elicited examples that serve as case-studies. We conclude with a summary of the forms we found, and where there were gaps, and what this means for how to think about the formal analysis of quantification and depicting signs in sign languages, and about symbolic abstraction in language more broadly.

1.1. Depicting signs/classifier predicates in sign languages

Classifier predicates, also known in the sign language literature as *depicting signs*, or *depicting classifiers*, have long been of interest in sign linguistics. Early work by Klima & Bellugi (1979) referenced classifier predicates and analyzed the handshape as a ‘pronoun’ and the movement the verb, which ultimately may not so far off the mark: the handshapes are based on semantic properties, and the movement and location are interpreted iconically (Emmorey & Herzig 2003). The entire construction has the distribution of a predicate, hence the name *classifiers* or *classifier predicates* (if we focus on the classifying properties of the handshape) or *depicting signs* or *depicting verbs* if we focus on their depictive properties. IDGloss, based at Gallaudet, focuses on their depictive properties, hence glossing these as DS (depicting sign) – here we will often gloss them as CL, but the two should be understood as notational variants. Figure 2 shows examples of different handshapes across 8 different sign languages, taken from Zwitserlood’s excellent overview of classifiers (Zwitserlood 2012).

One important thing to note connected to the terminology for sign language depicting signs/classifier predicates is that even though in this paper we will be discussing **classifiers** and **quantifiers**, the classifiers in question are entirely different from commonly studied quantizing classifiers in languages like Mandarin, in which a classifier has to be used to connect a noun phrase to a quantifier, all of which occurs in the nominal domain. In contrast, sign language classifiers are verbal classifiers, and do not seem to play

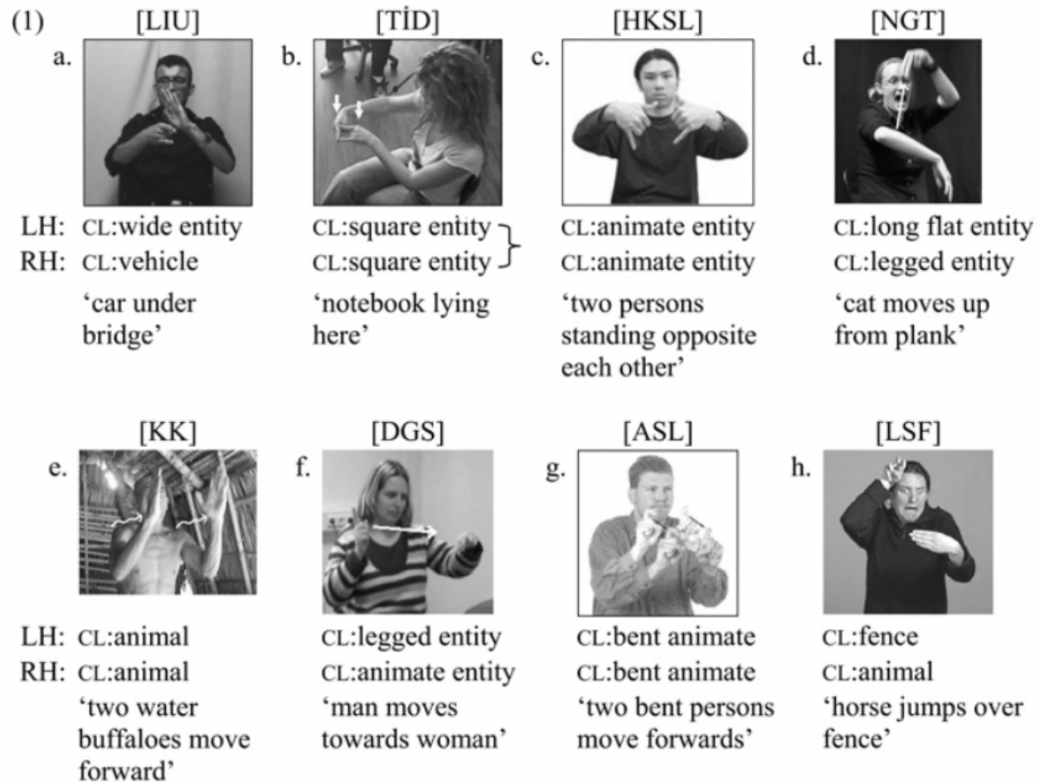


Figure 2: Classifier predicates across various sign languages, from Zwitserlood (2012). Sign language depicting classifiers function as predicates, with their handshape based on categorical semantic class that varies across sign languages, and their movement and location interpreted iconically

anything like the same kind of role in atomizing predicates for quantifiers.

Formal analyses of depicting classifiers take various forms. We build on the view by Davidson (2015) and Zucchi (2017) in which the handshape requires (encoded via a presupposition) that the entity the predicate applies to belongs to a particular semantic class. In the case of the classifier in Figure 3, this would be a chunky sized object. In contrast, the particular location and movement (in Figure 3, the exact thickness) are analyzed as basically a “co-sign gesture”, where the amount is taken as a demonstration that we relate to another eventuality, in this case one in which the book is a (theme) participant. Critically, the classifier sign involves a depictive component (the gesture) integrated into the structure of the predicate such that the entire predicate can depict and is understood in part through a highly iconic mapping (i.e. not purely through symbolic abstraction). In this example in Fig 3, the property depicted through this classifier predicate (being around the given thickness) is predicated of the sentence topic, BOOK, so that they sentence conveys that the book is of that particular (depicted) thickness.

Depicting signs are thus entirely syntactically integrated in some sense, because they play the role of a predicate in a clause (here, predicating something of the topic), and occur in the same modality as the rest of the highly abstract content of the sign language. At the same time, they express iconic/depictive content, similar to co-speech gestures. This dual status leads to an ideal testing ground for our question: what happens when such predicates interact with more abstract aspects of sign languages such as quantifiers? In the next section, we overview prior discussion of quantification in American Sign Language.

Handshape: semantic class, movement/location: demonstration event
 Entire DS_C(thick) a predicate that relates a participant and an eventuality



Figure 3: Modeling both symbolic and iconic aspects of sign language classifier predicates

1.2. Quantification

At the other end of the abstraction spectrum from depicting signs, ASL -like other sign languages- makes use of quantificational strategies that range across the same kinds seen in spoken languages, including examples of both adverbial and determiner quantifiers, as argued by Petronio (1995) and Abner & Wilbur (2017) and others. These authors pointed out many interesting ways to consider quantifiers in sign languages, such as information structural-influenced word order, distributivity, etc. Importantly, quantifiers of some sort seem to occur in all sign languages that have been studied seen so far, including new sign languages like Nicaraguan sign language (Kocab et al. 2022).

In a handbook comparing and contrasting quantificational strategies across languages, Abner & Wilbur (2017) and Kimmelman (2017) overview many strategies of quantification in sign languages, taking as their starting point the book's survey which is designed for spoken languages. This can be very helpful in many ways especially in leading to interesting comparisons between sign and spoken languages. However, a focus on comparison often means missing interesting data that wouldn't arise in spoken languages. For example, one might ask how nonmanuals play a role, such as eyebrow raising, and/or eyegaze? Or, how do quantifiers and the use of space for depiction and anaphora in sign languages interact? There are several reasons to think these questions are important, and of course they would not naturally arise for spoken languages given their modality-specific angles.

Along these lines, interesting observation is the relationship between quantifiers and their restrictor: a quantifier can be signed in the same position as its restrictor. Davidson & Gagne (2022) analyze this co-location of a quantifier and a plural locus as a type of partitive construction, e.g. 'one of those', and Saral & Kelepir (2020) argue the same for Turkish sign language, where co-location of the universal quantifier and a plural locus is interpreted as 'all of those' (Figure 4).

The use of space as the restrictor of a quantifier is one sign language specific aspect of quantification. Another comes from an observation involving binding: Graf & Abner (2012) observe that spatially associated pronouns turns out not to be possible with negative quantifiers, in contrast to non-negative quantifiers, even in minimal pairs which are otherwise well-formed when there are no spatially associated pronouns:

- a. EACH POLITICS PERSON_i TELL-STORY (IX_i) WANT WIN
Each politician said he wants to win.
- b. NO⁰ POLITICS PERSON_i TELL-STORY (*IX_i) WANT WIN
No politician said he wants to win.

Here, a positive quantifier (EACH POLITICS PERSON) appears to enter into a binding relationship with a pronoun (with singular morphology) in (a); but when we substitute a negative quantifier (NO POLITICS PERSON), suddenly the same kind of binding relationship appears to be degraded, and the sentence is

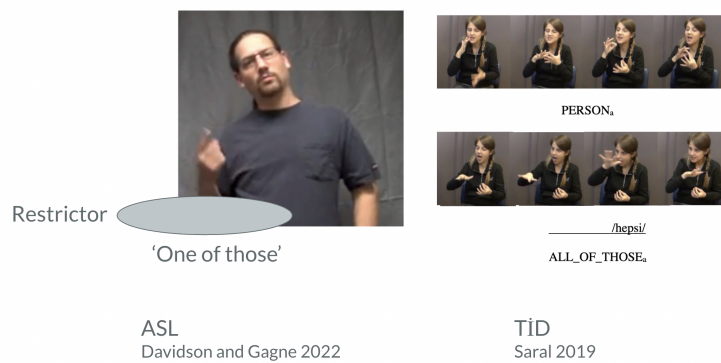


Figure 4: Quantifiers and space in sign languages (here, American sign language and Turkish sign language)

preferred to be expressed with an implicit pronoun (American Sign Language is a pro-drop language, Sandler & Lillo-Martin (2006)). Already, this raises the important point of what role the use of this space has and how and why it might compose with quantifiers.

In short, despite quite a lot of interest within formal semantics in how quantification works in sign languages, we observe that there is still a lot of groundwork to be done. In particular, existing work has done an excellent job of comparing sign language quantifiers to spoken language quantifiers, notably in two handbooks which set out to do precisely that: one classic (Petronio 1995), one recent (Kimmelman 2017, Abner & Wilbur 2017), both foundational and important when it comes to the typology of quantifiers in spoken language. But a tradeoff that comes with this focus on comparison between sign and speech is that it means less focus in areas sign languages might be uniquely well set up for, where a question might not even arise in spoken languages because they involve aspects of language that we do not find in the same way in spoken languages, such as loci discussed above, and integrated depicting predicates as discussed before that. In general, then, for this project we considered that eliciting natural productions for expressing quantification from signers is a good place to start, and then these can be analyzed for both more and less familiar aspects, instead of first searching for various kinds of quantification that we know exist in spoken language. In what follows, we use a naturally elicited production approach, and focus our analysis on one of these less well-studied areas, namely the integration of quantifiers with depicting predicates.

1.3. How quantification and depicting predicates interact

To orient our question, let us consider first the example from a short ASL narrative in Figure 5. This example was constructed between a linguist and a signer, so it was not entirely naturally occurring as ours will be, but it nicely illustrates the phenomenon we are interested in: on the one hand, an iconic use of gesture/space through a depicting classifier that depicts the arrangement of students behind a table, and then a negative quantifier NONEsym 'none', which then seems to say something about those depicted students, namely the students are the restriction for the quantifier ('none of them remember a card'). In Figure 5, in red is the depicting classifier of interest, and in blue is the quantificational structure. Note that in this case, the two are connected with an overt pronoun IXarc that points back to the referent that was depicted (the students standing in line), and the quantifier follows the pronoun.

It seems at first glance then that less abstract elements like iconic depictions interact highly with more abstract elements like quantifiers in sign languages. However, if we consider this example more closely, we might want to separate these elements into two separate clauses, in fact, with a clause break between the depicting classifier and the pronoun, where the line breaks. In terms of semantics, we might imagine the critical role of connecting the classifier and the quantifier in composition being played by the pronoun IXarc (them), which refers to the students introduced in the prior clause, but where there is no direct composition between the depiction and the quantifier. Could they perhaps compose directly? Thus, we set up our task to gather more data on depiction and quantifiers and the syntactic, semantic, and

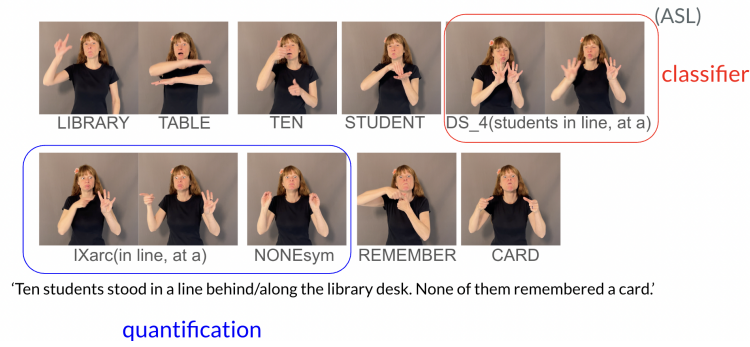


Figure 5: Integrating non-symbolic/iconic gestures and space into the same signal as syntactically structured symbolic language through sign language narrative



Figure 6: Elicitation stimulus example: a pair of AI-generated pictures where a target restrictor set (here, people) is difficult to easily count, and where the pictures differed in quantificational strength

information structural relationships between them. In particular, we ask what restrictions language might impose on the relationship between depiction and quantification, and focusing on how quantifiers interact with their restrictors in depicting contexts, and the role of information structure and anaphora in mediating this relationship.

2. Methodology

Our methodology involved conducting interviews with three native signers of American Sign Language over Zoom and recording their descriptions prompted by stimuli we created. The goal of our stimuli was to elicit quantificational structures in order to understand the role that information structure plays in relating depictions to quantifiers. As such, we roughly built on a methodology that was successful in a prior study eliciting lexical/classical Aristotelian quantifiers in Nicaraguan Sign Language by Kocab et al. (2022). One insight provided by that study was that single picture stimuli alone are unlikely to elicit quantifiers because for a single picture signers will focus on expressions of quantity that are highly depictive in some details (location and arrangement, for example), but may not necessarily think to explicitly express information about quantity. To especially elicit expressions of quantity, a better method is paired contrasts (e.g. some vs. none, or all vs. some) in picture sequences, which do successfully elicit classical quantifiers of the sort that we are interested in. The idea is that pictures will be similar in many respects except for one salient quantity difference between them. Another insight provided by that study is that our pictures used a high number of objects for the target quantification (e.g. more than 4) to minimize the ability to provide an exact number instead (Feigenson et al. 2004). Figure 6 shows one of our elicitation prompt contexts in which there are many people (too many to easily count) in the first picture, and then in the second picture they have all disappeared. This contrast in the second photo can naturally elicit either universal quantifiers (*all disappeared*) or negative quantifiers (*there are none*).

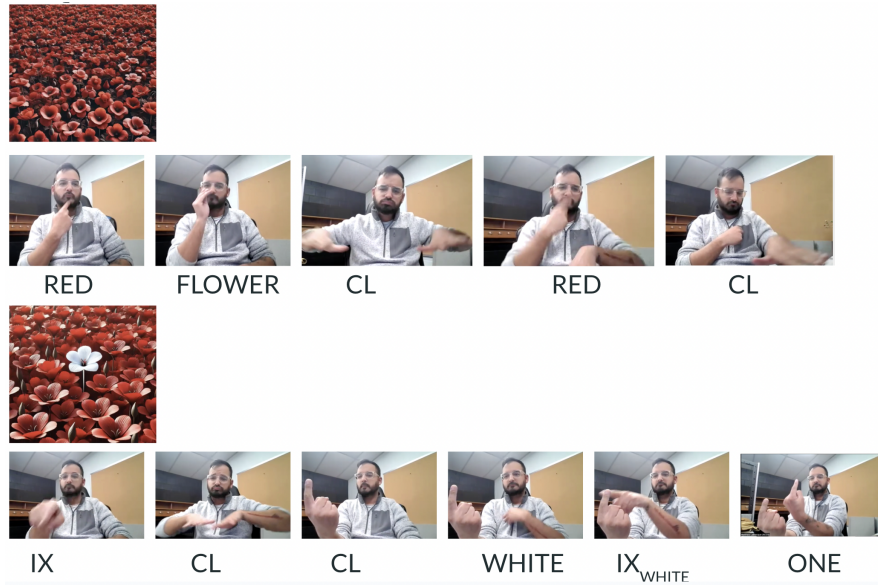


Figure 7: Backgrounded topic, followed by new information focus

3. Information structure, depiction, and quantification

3.1. Topic and comment

We first present examples from our data set that highlight several aspects of information structure and the use of space in sign languages that interact with quantifiers, to show the success of this method in eliciting the kind of structures we are interested in. Consider Figure 7: the first picture shows a large number of very similar looking red flowers covering the frame. To describe this picture, the signer sets up the sentence topic RED FLOWER and then uses a depicting classifier to convey the area covered by these red flowers. The second picture looks similar except there is one white flowers emerging among the red flowers. For this frame, the signer begins with an indexical pointing sign IX and a classifier that depicts the background similar to the previous frame, and then uses a classifier for a tall narrow object (the flower stem) to depict the single flower in focus, describes it as WHITE, and then uses the number quantifier ONE. Here the depicting classifier for the background is shared across the frames and comes first (as a kind of scene setting topic) in the description of the second frame, while the contrast between the two frames is the one white flower, which is mentioned after the topic, the focus of the sentence.

It has long been noted that when sign languages recruit space for iconic purposes, given/backgrounded information tends to come first, followed by new/foregrounded information (Kimmelman & Pfau 2021), and indeed this is what happens in the example in Figure 7. The quantifier occurs outside the topic here, as it generally does in these structures in ASL and in other sign languages.

3.2. Contrast

Another important mechanism for conveying information structure in sign languages is the use of left and right signing space to convey contrast: Navarrete-González (2023) provides a detailed examination of this use of contrast in coordinating structures in Catalan Sign Language, and similar uses of space for contrast can be found in ASL. One can certainly also recognize this use of space both in sign languages and in gestures accompanying spoken language making use of the metaphor DISTANCE-IS-DIFFERENCE. Our stimuli elicited this structure as well: Figure 8 shows that the stimuli involve a clear contrast in color: the first presented frame shows all red flowers, while the second frame shows all white flowers. Thus, the layout is roughly the same, but the contrast is a natural point to highlight, and the signer does this through associating the two frames to two distinct locations in signing space: one on the left side (with the red

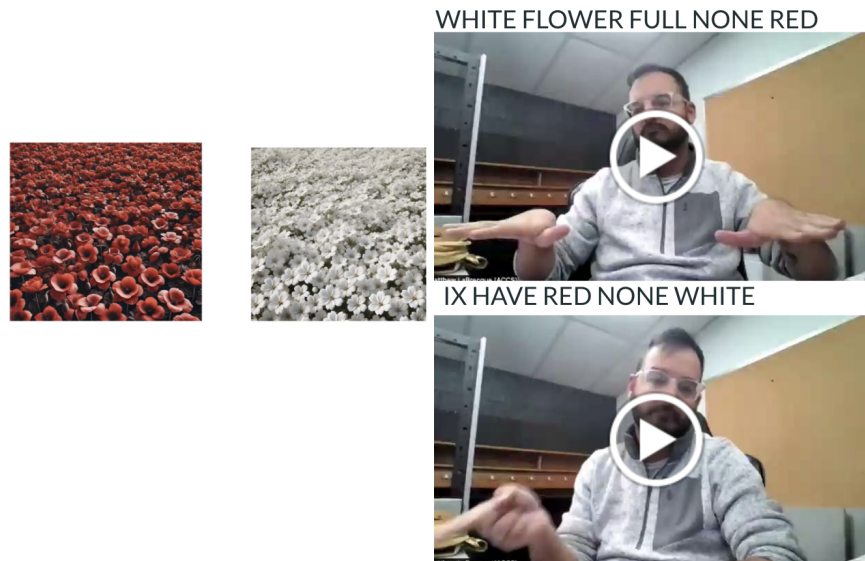


Figure 8: The use of left and right signing space for contrast: here, the contrasting colors between the first and second frames

flowers), and one on the right side (with the white flowers).

In this example in Figure 8, the signer also used a lexical quantifier *NONE*, and predicates of the flowers that none were red, and predicates of the red flowers that none were white. Here, two sides of signing space are used for the two situations, and the scene is established as a topic; the quantified phrase (e.g. *NONE RED* or *NONE WHITE*) follows.

We can see a similar use of space in another example in Figure 9, where the two frames contrast in the number of fish, and are also placed in the left and right of the signing space in a way that conveys that they contrast. These examples also both highlight a question that we mentioned in our introduction: how might a negative quantifier in particular interact with depicting elements? Recall some prior work on quantifiers in sign languages have observed that negative quantifiers show a different pattern than positive quantifiers when it comes to iconic content, in particular to binding loci (Graf & Abner 2012). We see in both Figures 8 and 9 that the sign *NONE* is used, which is perhaps surprising given this background, and so in the next section focus our investigation specifically on negative quantifiers to address questions raised in prior literature about negative quantification and iconicity.

4. Negative quantifiers and iconic depictions

Overall, our examples elicited many cases of a quantifier immediately followed by a lexical sign, such as *PEOPLE*, *FISH*) shown in Figure 10, and *RED*, *WHITE* earlier in Fig 8. In this section, we focus on these examples to understand how the distribution with respect to quantifiers (information structural, semantic, and syntactic) compare and contrast between lexical and depicting signs.

In some examples, the lexical sign following the quantifier has the information structural role of presenting new information. In these examples, the restrictor set seems to be implicit or introduced as a topic. We can see this in the case from Fig 10, where *WHITE FLOWER FULL* introduces the topic (white flowers) which also acts as the (implicit) restrictor of the quantifier for the next clause *NONE RED*, with *NONE* as the quantifier, and *RED* as its scope.

In other examples, the negative quantifier combines with a lexical predicate in order to state non-existence, as in *NONE PEOPLE*, or *NONE FISH* in Figure 10. In these examples, the lexical sign is used with something like a *thetic* information structure, i.e. with all information equally new/equally backgrounded. Of interest is our observation that beyond examples where we see a lexical sign as the topic restrictor, or in a *thetic* sentence, we did not elicit any clear examples of English-like word order of *QUANTIFIER-*



Figure 9: The use of left and right signing space for contrast: here, the contrasting amounts of fish

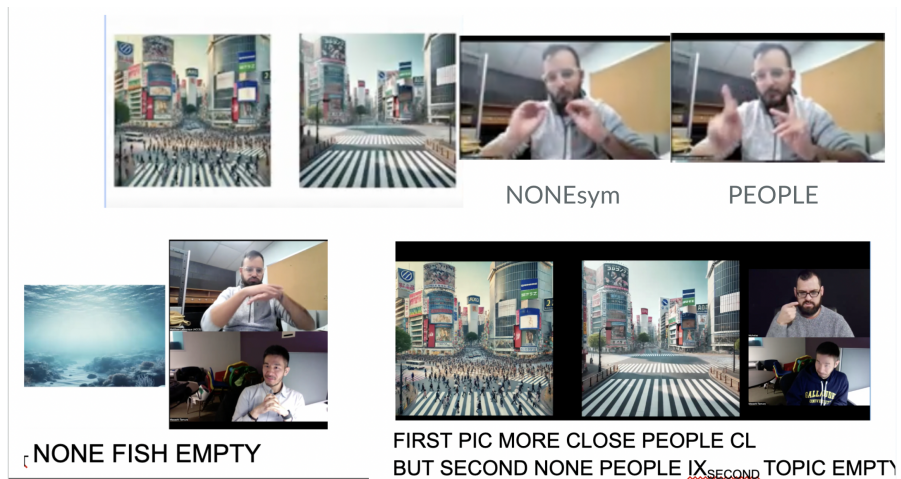


Figure 10: The negative quantifier NONE in context

RESTRICTOR-SCOPE for negative quantifiers, e.g. *NONE FISH SWIM*. The utterance *NONE FISH EMPTY* has this word order, but it is not understood as QUANTIFIER-RESTRICTOR-SCOPE (which would be interpreted as 'none of the fish are empty'); rather it is athetic *NONE FISH*, followed by an emphatic *EMPTY* ('There are no fish, it is empty').

For comparison, we can also ask what happens when we go beyond lexical signs to depicting signs. Depicting classifier predicates are typically analyzed as having the same semantic type as these kind of lexical nouns and adjectives, e.g. as a predicate (a function mapping individuals to true values), so from the perspective of compositional semantics we might expect them to combine with a quantifier in exactly the same way. However, we interestingly observe more constraints in the way that depictions interact with negative quantifiers: they can be used before the negative quantifier to describe a possible scene, but in these cases always precede the quantifier. An example can be seen in Figure 10, where the classifier predicate is used to describe the busy scene of people (*PEOPLE CL*) in the first clause, in contrast to the negative quantifier in the second clause (*NONE PEOPLE*). What we never find is a depicting classifier predicate immediately following a negative quantifier, e.g. *NONE CL*. We might expect this to be a reasonable thetic sentence exactly of the kind we see like *NONE PEOPLE* or *NONE FISH*, but despite pictures that naturally elicit depicting quantifiers, we have no examples of *NONE CL*.

We might ask more broadly about classifiers and expressions of no quantity/negation. In general, two strategies seem to emerge to express negative quantification in our contexts. One strategy, as we have noted, involves a negative quantifier (e.g. *NONE*) followed by a lexical restrictor, such as *PEOPLE* and is familiar from spoken languages. The second involves a negative expression like *CL(EMPTY)* which can predicate a property (e.g. emptiness) of an area, and can be signed in the area where it applies. These are directly compared in Figure 11.

- **First strategy.** A depicting classifier sets up the relevant region and **CL(empty)** predicates emptiness of that region.
- **Second strategy.** A **negative quantifier** (*NONE*) combines with a **lexical restrictor** (*PEOPLE*), yielding a standard quantificational NP.

We can similarly see these strategies contrasted in Figure 12, where again the emptiness of a small candy bag is depicted using a classifier sign in one strategy; in another (lexical quantifier) strategy, the same signer uses a sign for *NONE* (with the lexical sign unpronounced, the topic *CANDY* in the discourse).

- **First strategy.** As in Figure 11–1, a depictive use of space establishes the locus and **CL(empty)** predicates absence (e.g., an empty container/area).
- **Second strategy.** A **negative quantifier** appears **without an overt lexical noun** (here, a lexical item *CANDY* pronounced earlier in the story). Instead, the restrictor is supplied anaphorically by a demonstrative that points back to the previously depicted set.

Taken together, Figures 11 and 12 illustrate two strategies of negative quantification, which may either (i) predicate absence iconically via *CL(empty)* at a spatial locus, or (ii) employ a manual negative quantifier that composes with a lexical predicate or over an anaphorically retrieved predicate when a lexical noun is absent. This contrast underscores that depicting classifiers can appear in quantificational contexts but, along with prior literature, suggests there may be an additional constraint involving iconic depiction and quantification, especially negative quantification.

Although he was focused on different kind of data involving quantification (especially: concord structures such as negative and distributive concord), Kuhn (2020) observes a similarly surprising distribution involving negative quantifiers in sign languages and suggests that the anaphora system in sign languages may be constrained by a bias that something placed in signing space exists. In particular, Kuhn suggests that an “iconic inference of existence” may be the reason there is not negative concord between manual signs in sign languages (except for a few exceptions that may prove the rule, e.g. Russian SL). We find the idea that the use of space is somewhat inherently iconic, and thus resistant to negative quantification, to resonate with our entirely different data set, where depicting classifiers had a more restricted distribution than lexical signs when it comes to negative quantifiers.

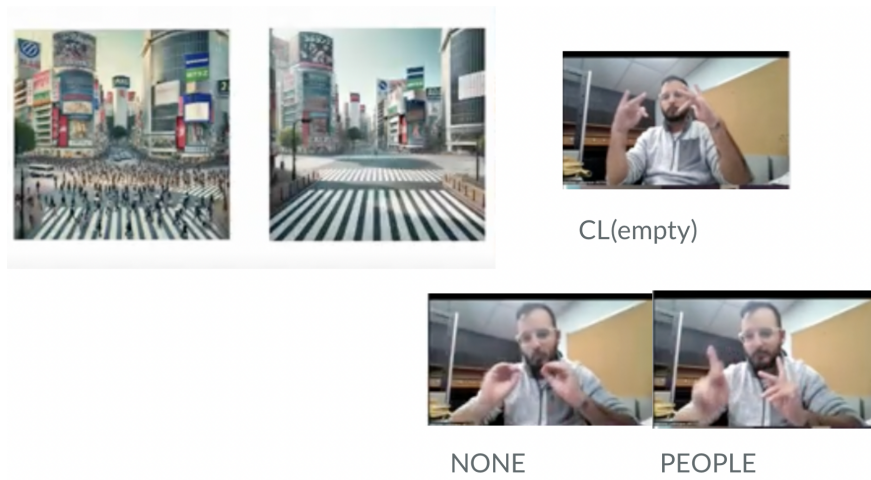


Figure 11: Two strategies for negative quantification: a negative expression like CL(empty), and a negative quantifier and a lexical restrictor

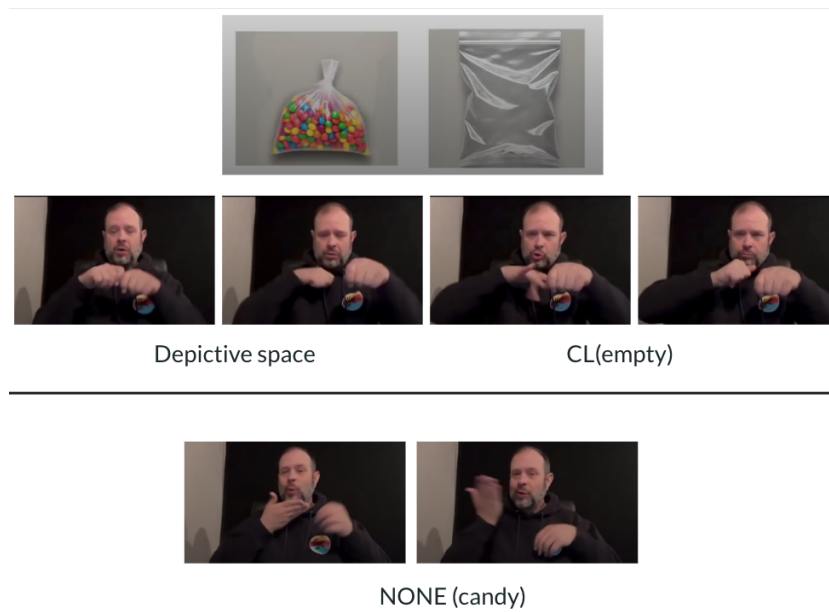


Figure 12: Two strategies for expressing absence: a depictive use of space combined with CL(empty), and a lexical negative quantifier like NONE(candy).

In fact, they also fit with a third observation, along with ours and Kuhn's, where negative quantifiers seem to be especially restricted with iconic signs. Recall the observation we mentioned above from Graf & Abner (2012) that negative quantifiers do not participate in binding relationships with pronominals:

- a. EACH POLITICS PERSON_i TELL-STORY (IX_i) WANT WIN
Each politician said he wants to win.
- b. NO⁰ POLITICS PERSON_i TELL-STORY (*IX_i) WANT WIN
No politician said he wants to win.

Here is yet another case where negative quantifiers show restrictions with iconic signs, in this case the spatial use of loci for sign language anaphora. There are differences, for sure: their example involves quantification via binding a pronoun (IX) ranging over individuals, which did not occur in our data naturally. This may reflect an underlying bias in the language, but it could also be an influence of our stimuli, which did not naturally prompt embedded clauses/attitude reports of this kind, or relative clause structures, since it's worth noting that such bound individual examples in sign languages tend to involve clausal embedding. In our data when it comes to examples involving negative quantifiers and our stimuli, they tended to have the structure of a topicalized restrictor, then the quantifier and its scope. Still, nevertheless an interesting pattern emerged: overt quantificational structures especially with negative quantifiers involved lexical predicates, despite our stimuli strongly favoring depicting classifier predicates in initial setup.

Recall, from a purely computational/symbolic perspective we might expect that any depicting classifier could participate in a quantificational structure just as any lexical predicate: both Davidson (2015) and Zucchi (2017) model the semantics of a depicting predicate as having the same compositional type as any other predicate over individuals, e.g. type $\langle e, t \rangle$, so all else being equal we might expect it to be able to combine with a quantifier just like any other of this type, e.g. a relative clause. However, we found no examples of depicting classifiers used in this way, despite our stimuli eliciting many depicting classifiers in descriptions outside of the negative quantificational structure.

Thus, we provide a new source of empirical support based on different kinds of structures including thematic sentences, and a different elicitation methodology that favored natural productions involving iconic depiction, for the general observation that sign languages seem to resist negation, especially negative quantifiers, in composition with iconic structures. We tentatively suggest a way to capture these observations through the use of an (implicit) pronoun picking out the topic, which can be introduced with a depicting classifier. Consider, for example, the examples in Figure 11: a classifier predicate is originally used to depict the people moving around the busy square, setting up this scene as the discourse topic. One continuation CL(empty) simply predicates emptiness of this situation/scene. The other, NONE PEOPLE, also predicates the lack of people of this situation/scene via an implicit pronoun (known to be licensed in ASL), e.g. that it (that scene) has no people.

5. Discussion and Conclusions

In this paper, we have examined how two extremes on the spectrum of abstraction in sign language—depicting predicates and quantifiers—interact. Depiction represents the least abstract end of the spectrum, understood through a relatively direct mapping from world to form. Quantification, by contrast, exemplifies the most abstract end, involving relations between two abstractions (sets or functions).

Our data provide further support, of a new kind, for the generalization that these two extremes do not compose directly. Instead, they appear to interact in limited ways. For example, a classifier can depict and thus establish a sentence topic, and a (overt or cover) pronoun can then participate in predication or quantificational structures. But, we do not see classifier predicates used with the same distribution in quantificational structures, especially negative quantificational structures, as lexical predicates. This restriction is particularly striking given recent arguments that pictures, gestures, and other non-lexical content can contribute meaning with the same compositional potential as symbolic language (Tieu et al. 2019). The fact that sign language classifiers cannot freely combine with quantifiers in the same way as lexical predicates indicates that there may be principled constraints on how depiction and quantification interact. Of course, our study has many limitations, most notably that it only involves positive data given these were all natural productions; future studies should probe variants for ungrammaticality

judgements to test these predictions.

Looking forward, our findings point to the need for further investigation of information-structural factors in shaping the interaction between depiction and quantification, both within ASL and cross-linguistically across sign languages. Such work promises to shed new light on the nature of symbolic abstraction in human language more broadly, by clarifying how highly iconic content and highly abstract operators can be integrated within a single linguistic system.

References

- Abner, Natasha & Ronnie B Wilbur. 2017. Quantification in american sign language. In *Handbook of quantifiers in natural language: volume ii*, 21–59. Springer.
- Davidson, Kathryn. 2015. Quotation, demonstration, and iconicity. *Linguistics and philosophy* 38(6). 477–520.
- Davidson, Kathryn & Deanna Gagne. 2022. “more is up” for domain restriction in asl. *Semantics and Pragmatics* 15. 1–1.
- Emmorey, Karen & Melissa Herzig. 2003. Categorical versus gradient properties of classifier constructions in asl. In *Perspectives on classifier constructions in sign languages*, 231–256. Psychology Press.
- Feigenson, Lisa, Stanislas Dehaene & Elizabeth Spelke. 2004. Core systems of number. *Trends in cognitive sciences* 8(7). 307–314.
- Graf, Thomas & Natasha Abner. 2012. Is syntactic binding rational? In *Proceedings of the 11th international workshop on tree adjoining grammars and related formalisms (tag+ 11)*, 189–197.
- Kendon, Adam. 2004. *Gesture: visible action as utterance*. Cambridge University Press.
- Kimmelman, Vadim. 2017. Quantifiers in russian sign language. In *Handbook of quantifiers in natural language: volume ii*, 803–855. Springer.
- Kimmelman, Vadim & Roland Pfau. 2021. Information structure: theoretical perspectives. In *The routledge handbook of theoretical and experimental sign language research*, 591–613. Routledge.
- Kita, Sotaro. 1997. Two-dimensional semantic analysis of japanese mimetics.
- Klima, Edward S & Ursula Bellugi. 1979. *The signs of language*. Harvard University Press.
- Kocab, Annemarie, Kathryn Davidson & Jesse Snedeker. 2022. The emergence of natural language quantification. *Cognitive Science* 46(2). e13097.
- Kuhn, Jeremy. 2020. Logical meaning in space: iconic biases on quantification in sign languages. *Language* 96(4). e320–e343.
- McNeill, David. 2016. *Why we gesture: the surprising role of hand movements in communication*. Cambridge University Press.
- Navarrete-González, Alexandra. 2023. Focus and contrast in catalan sign language (lsc): form and interpretation. *Sign Language & Linguistics* 26(1). 149–158.
- Petronio, Karen. 1995. Bare noun phrases, verbs and quantification in asl. In *Quantification in natural languages*, 603–618. Springer.
- Quer, Josep. 2012. Quantificational strategies across language modalities. In *Logic, language and meaning: 18th amsterdam colloquium, amsterdam, the netherlands, december 19-21, 2011, revised selected papers*, 82–91.
- Sandler, Wendy & Diane Carolyn Lillo-Martin. 2006. *Sign language and linguistic universals*. Cambridge University Press.
- Saral, Burcu & Meltem Kelepir. 2020. The universal quantifier ‘all’ in turkish sign language. In *Morphological complexity within and across boundaries*, 353–384. John Benjamins Publishing Company.
- Schlenker, Philippe. 2018. Gesture projection and cosuppositions. *Linguistics and Philosophy* 41(3). 295–365.
- Tieu, Lyn, Philippe Schlenker & Emmanuel Chemla. 2019. Linguistic inferences without words. *Proceedings of the National Academy of Sciences* 116(20). 9796–9801.
- Zucchi, Sandro. 2017. Event categorization in sign languages. In *Handbook of categorization in cognitive science*, 377–396. Elsevier.
- Zwitserslood, Inge. 2012. Classifiers. In *Sign language: an international handbook*. de Gruyter.